

DNS Security Reading Assignment

Due Date: 6th April'26

Submission: Submit your answers as a single PDF document.

This assignment requires you to analyze recent, high-profile DNS security incidents. The provided readings detail specific vulnerabilities, active attack campaigns, and fundamental weaknesses in the DNS ecosystem. Your responses must be thorough, analytical, and directly reference the technical details from the assigned material.

Assigned Readings

Reading resource is given in [excel](#).

Questions

Answer all five questions below. Each answer should be a minimum of two substantial paragraphs.

Q1. The BIND 9 Vulnerability (CVE-2025-40778)

Explain the root cause of the BIND 9 vulnerability (CVE-2025-40778). How does it violate fundamental DNS trust assumptions, and why does it make cache poisoning significantly easier compared to traditional attacks?

Q2. The Detour Dog Campaign

Describe the complete attack chain used in the Detour Dog campaign. Specifically explain how DNS is used not just for resolution, but as an active component in command-and-control (C2) and malware delivery.

Q3. Comparison of DNS Exploitation Methods

Using examples from the readings, compare two different ways DNS is exploited:

- **DNS as a target** (e.g., cache poisoning, hijacking)
- **DNS as an attack infrastructure** (e.g., malware delivery, C2)

Explain how the attacker goals and techniques differ in both cases.

Q4. Weak DNS Governance and Domain Hijacking

Based on the domain hijacking discussion, explain how weak DNS governance (registrar access, misconfigurations, expired domains) can lead to large-scale attacks. Why are such attacks often harder to detect compared to traditional network attacks?

Q5. Persistent Weaknesses in DNS

Despite decades of improvements (randomization, DNSSEC, etc.), DNS continues to be exploited in modern attacks. Using examples from all the readings, analyze at least three fundamental reasons why DNS remains a weak point in the internet ecosystem.